Variability of Subjective Classifications of Corneal Topography Maps From LASIK Candidates

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ABSTRACT

PURPOSE: To evaluate the variability of subjective corneal topography map classification between different experienced examiners and the impact of changing from an absolute to a normative scale on the classifications.

METHODS: Preoperative axial curvature maps using Scheimpflug imaging obtained with the Pentacam HR (Oculus Optikgeräte, Wetzlar, Germany) and clinical parameters were sent to 11 corneal topography specialists for subjective classification according to the Ectasia Risk Scoring System. The study population included two groups: 11 eyes that developed ectasia after LASIK and 14 eyes that had successful and stable LASIK outcomes. Each case was first reviewed using the absolute scale masked to the patient group. After 3 months, the same cases were represented using a normative scale and reviewed again by the same examiners for new classifications masked to the patient group.

RESULTS: Using the absolute scale, 17 of 25 (68%) cases had variations on the classifications from 0 to 4 for the same eye across examiners, and the overall agreement with the mode was 60%. Using the normative scale, the classifications from 11 of 25 (44%) cases varied from 0 to 4 for the same eye across examiners, and the overall agreement with the mode was 61%. Eight examiners (73%) reported statistically higher scores ($P < .05$) when using the normative scale. Considering all 550 topographic analyses (25 cases, 11 examiners, and two scales), the same classification from the two scales was reported for 121 case pairs (44%).

CONCLUSION: There was significant inter-observer variability in the subjective classifications using the same scale, and significant intra-observer variability between scales. Changing from an absolute to a normative scale increased the scores on the classifications by the same examiner, but significant inter-observer variability in the subjective interpretation of the maps still persisted.

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Computerized videokeratography (corneal topography) emerged as an important complementary examination in ophthalmology. With the popularity of refractive surgery using excimer laser ablation in the 1990s, corneal topography became an essential tool to identify preoperative patterns that would indicate a less favorable prognosis and risk of postoperative corneal ectasia.[1-6] Specific indices to detect corneal abnormalities were developed by Maeda et al.[7,8] and Rabinowitz.[9] In these cases, subjective analysis of the topography by the investigator to determine topographic pattern combined with patient-specific clinical information defined the diagnoses.

In 2008, the Ectasia Risk Scoring System (ERSS) was published by Randleman et al.[10] based on a retrospective case-control study that evaluated Placido-based corneal topography, central corneal thickness, the degree of preoperative myopia, residual stromal bed thickness, and patient age in a weighted fashion. As part of the ERSS, a topographic pattern scoring system was also defined. This study and a follow-up validation

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study by Randleman et al.[11] confirmed abnormal preoperative topography as the most significant predictive variable. However, the ERSS had a false-negative rate of 4% to 8% and a false-positive rate of 6%.[10,11] Other studies have reported relatively high false-negative rates,[12] but a higher number of false-positive results has been reported. In addition, lower true positive rates were also reported,[13,14] although none of these publications included patient topographies for review or analysis.

The purpose of this study was to evaluate variability of corneal topographic pattern scores generated from the Scheimpflug images obtained with the Pentacam HR (Oculus Optikgeräte, Wetzlar, Germany) based on ERSS criteria among a group of experienced examiners using two separate color-coded scales (0.5 and 1.5 diopters [D]), using two groups of patients who did and did not develop postoperative corneal ectasia after LASIK with examiners blind to patient outcomes.

PATIENTS AND METHODS

Eleven eyes that developed ectasia after LASIK and 14 eyes with stable LASIK outcomes (follow-up > 18 months; mean: 19.8 months, range: 18 to 28 months) were selected for this study. The preoperative axial (or sagittal) curvature map of each case was obtained from corneal tomography (Pentacam HR). All examinations showed high rates of quality, including quality approval by the instrument software (quality specification). These maps and the clinical parameters (age, manifest refraction, visual acuity, central corneal thickness, and predicted residual stromal bed) were sent to 11 corneal topography specialists for masked subjective classification accordingly to the ERSS (0 = normal or symmetric bowtie; 1 = asymmetric bowtie; 3 = inferior steepening pattern or skewed radial axis; 4 = abnormal topography). Each case was first sent using the absolute Smolek–Klyce 1.5 D scale and then resent to the same examiners after a period of 3 months for a new classification using a 0.5 D normative scale (**Figures 1-2**) in a masked fashion.

Statistical analyses were performed using the Wilcoxon signed rank and Kruskal–Wallis tests to compare the differences between the examiners using the same scale and to compare the differences between the two scales by the same examiner.

RESULTS

In the first evaluation with the absolute 1.5 D scale, the variation of classifications from 0 to 4 across the examiners was observed in 17 to 25 cases (68%). The average agreement with the mode (number of examiners who choose the same classification as the mode [the value that appears most often in a set of data] in each case) was 60% among all cases (ranging from 36% to 100%). Only one case had 100% of agreement among all examiners (**Table 1**).

In the second evaluation with the normative 0.5 D scale, 11 of 25 cases (44%) had a classification variation from 0 to 4 across examiners. The average agreement with the mode was 61% among all cases (ranging from 45% to 91%). None of the cases had 100% agreement among all examiners (**Table 1**). The **Appendix** (available in the online version of this article) includes all results of the topography analysis by the examiners using 1.5 and 0.5 D scales, along with the agreement with the mode in circumstance.

The mode of classifications with the absolute Smolek–Klyce 1.5 D scale was statistically lower (Wilcoxon test, $P = .0033$) compared to the mode obtained with the normative 0.5 D scale. Considering all 275 classifications (11 examiners multiplied by 25 cases), statistical differences were found between the two scales (Wilcoxon test, $P < .0001$), with higher values on the normative 0.5 D scale (2.13 ± 1.56) compared with the absolute Smolek–Klyce 1.5 D scale (1.47 ± 1.55). Forty-four percent of the 275 analyses were exactly equal on the two scales, 42% had higher values on the normative 0.5 D scale, and 14% had higher values on the absolute Smolek–Klyce 1.5 D scale. Eight of 11 examiners (73%) had statistically higher classifications using the 0.5 D normative scale (Wilcoxon, $P < .05$).

DISCUSSION

This study assessed the variability of subjective assessment of corneal topographic pattern in 11 eyes that developed ectasia after LASIK and 14 eyes that underwent LASIK and were stable, as judged by experienced reviewers not informed of the patient outcomes during the review. The results demonstrate great variability of these subjective topographic classifications among and between examiners for both the normative 0.5 D and the American National Standards Institute standard absolute 1.5 D color scales.

The development of computerized corneal topography from the 1990s is closely linked to advances in corneal refractive surgery.[15] This topic has gained significant importance since the first reports of postoperative corneal ectasia,[16,17] and subsequent reported cases considered to be without preoperative risk factors for ectasia.[18,19] Thereafter, subjective evaluations of corneal topography performed preoperatively in these patients started to be questioned.[17] However, despite the emergence of new tests for the characterization of the cornea,[20-22] including anterior segment tomography,[23-27] epithelial thickness profiles,[28-31] and direct biomechanical measurements,[19,32,33] the correct

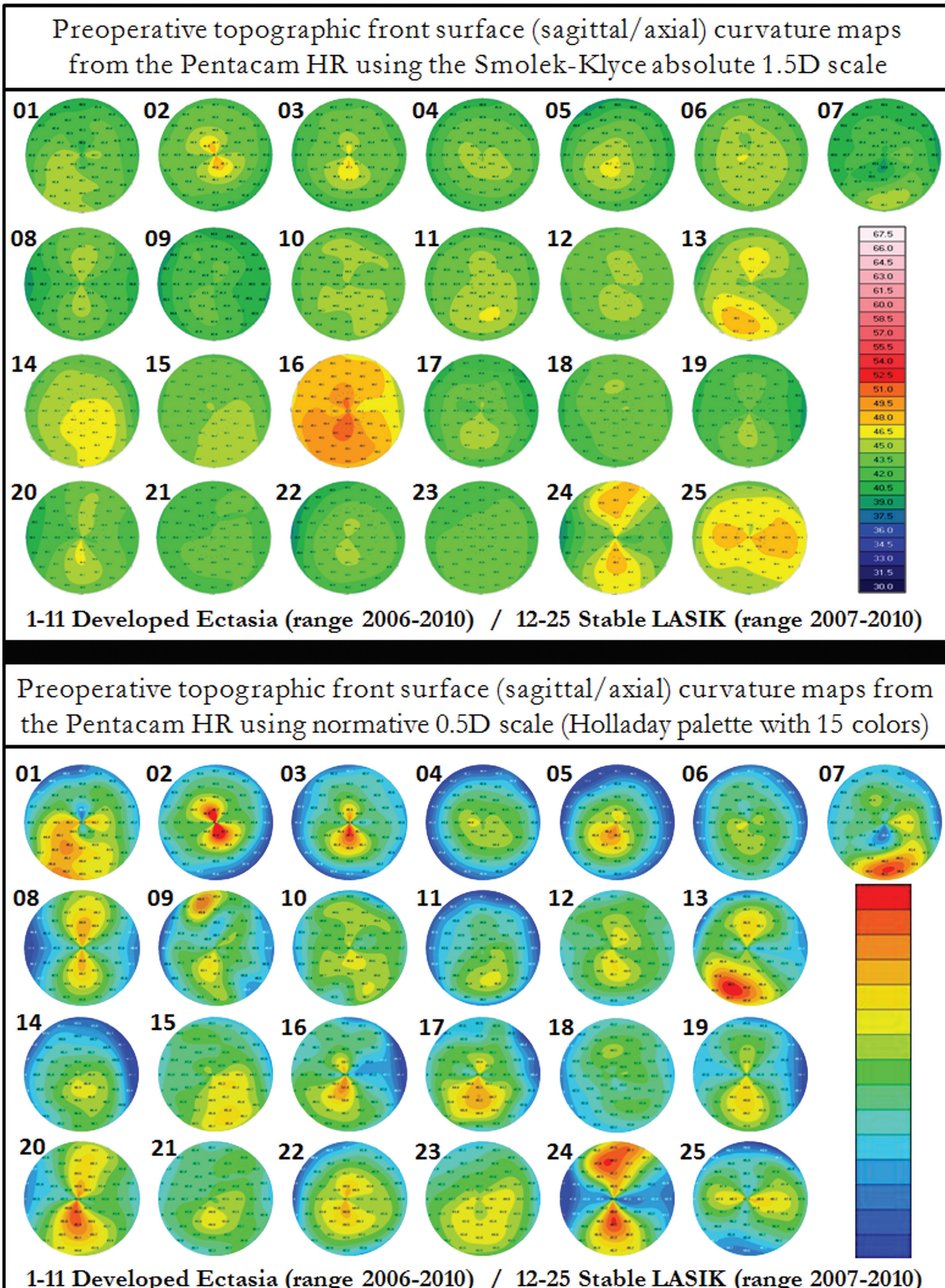

Figure 1. The 25 preoperative sagittal maps from Pentacam HR (Oculus Optikgeräte, Wetzlar, Germany) using the two scales.

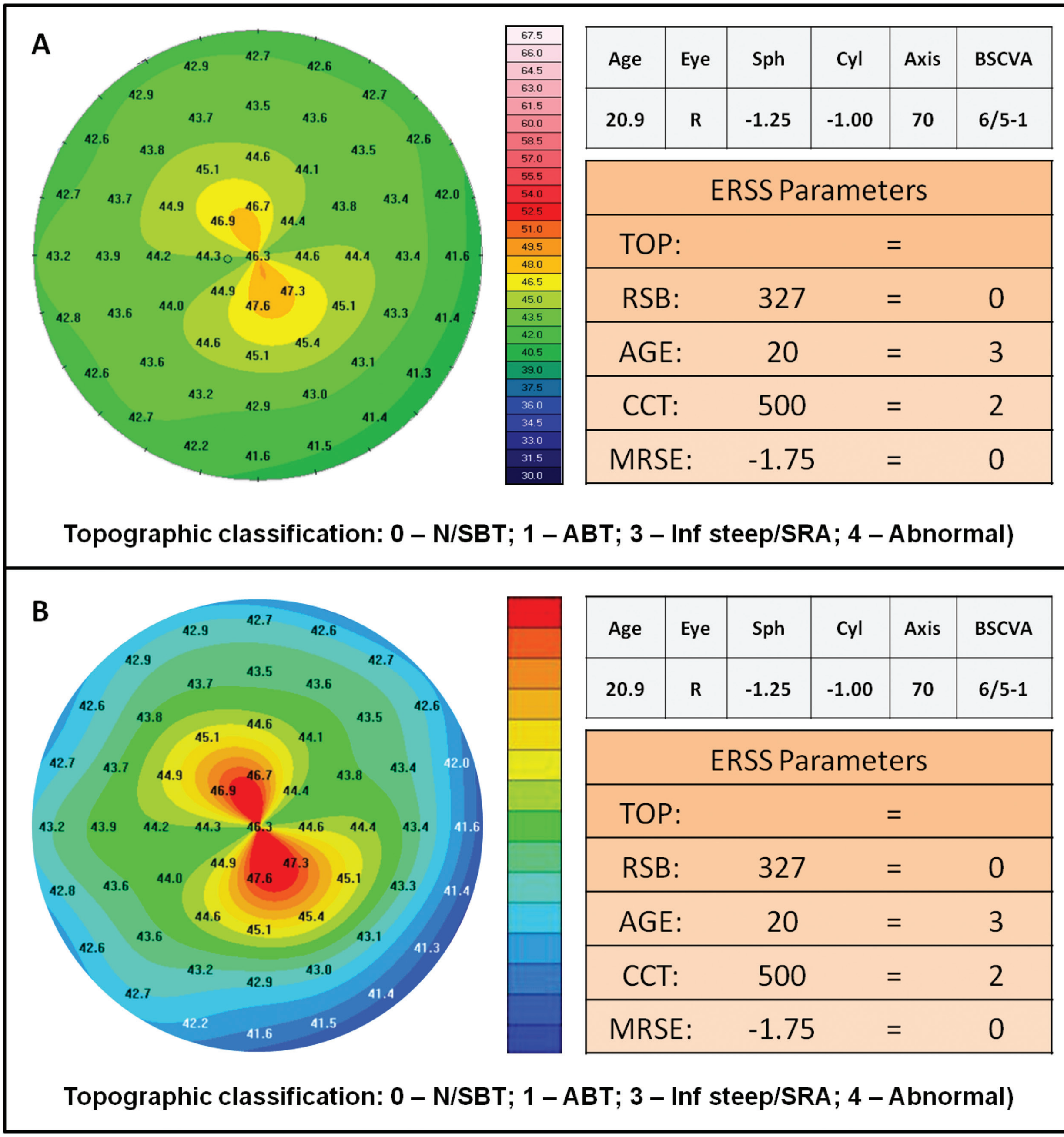

Figure 2. (A) Example of data sent for the first classification (with absolute 1.5 diopter scale). (B) Data of same case resent after 3 months to the new classification (with normative 0.5 diopter scale). ABT = asymmetric bow tie; BSCVA = best-spectacle corrected visual acuity; ERSS = Ectasia Risk Scoring System; CCT = central corneal thickness; MRSE = manifest refraction spherical equivalent; N/SBT = normal/symmetric bow tie; RSB = residual stromal bed; SRA = skewed radial axis; TOP = topographic classification

subjective evaluation of topographic patterns is still essential in the preoperative assessment of refractive surgery candidates.[34]

Retrospective studies involving careful evaluation of the preoperative corneal topography of patients who developed ectasia after LASIK have been performed.[18,35,36] Clinical parameters have been analyzed together to quantify the risk for developing ectasia in these eyes. However, it is important to emphasize that the subjective nature of the retrospective analysis becomes even more clouded when the reviewer knows the patient's diagnosis and outcome.

As hypothesized, changing the color scale from a 1.5 D step to a 0.5 D step scale had a significant impact

TABLE 1

Results of Topography Analysis (Classification 0, 1, 3, or 4) by the Examiners Using 1.5 and 0.5 Diopters (D) Scales and the Agreement Mode for Each Case

Case	Mode (Range) 1.5 D Scale	Mode (Range) 0.5 D Scale	Agreement With the Mode (%) 1.5 D Scale	Agreement With the Mode (%) 0.5 D Scale
01	3 (0–4)	4 (3–4)	45.45	72.73
02	4 (0–4)	4 (0–4)	45.45	54.55
03	1 (0–4)	4 (1–4)	54.55	54.55
04	0 (0–4)	0 (0–4)	63.64	72.73
05	4 (0–4)	3 (1–4)	36.36	54.55
06	0 (0–3)	0 (0–4)	63.64	45.45
07	3 (1–4)	4 (3–4)	45.45	81.82
08	0 (0–4)	0 (0–1)	90.91	72.73
09	0 (0–4)	1 (0–4)	45.45	45.45
10	0 (0–4)	0 (0–4)	72.73	54.55
11	1 (1–4)	3 (1–3)	45.45	90.91
12	0 (0–4)	3 (0–4)	45.45	54.55
13	4 (0–4)	4 (3–4)	72.73	90.91
14	3 (0–4)	3 (1–4)	54.55	54.55
15	1 (0–4)	3 (1–4)	72.73	54.55
16	4 (0–4)	4 (1–4)	36.36	54.55
17	3 (1–3)	3 (1–4)	54.55	45.45
18	0 (no range)	0 (0–4)	100.00	81.82
19	1 (0–3)	1 (1–3)	63.64	54.55
20	1 (0–4)	3 (0–4)	45.45	63.64
21	0 (0–1)	3 (0–4)	63.64	63.64
22	0 (0–4)	0 (0–4)	63.64	45.45
23	0 (0–1)	3 (0–4)	90.91	54.55
24	0 (0–4)	0 (0–3)	63.64	54.55
25	0 (0–4)	0 (0–3)	63.64	54.55
			Average = 60	Average = 61

on topographic pattern scores. Although the numbers on each topography map did not change with the change of scale, the color pattern differences allowed reviewers to more easily notice suspicious characteristics with the normative 0.5 D scale. The change in scale resulted in 8 of 11 examiners reporting statistically higher (more abnormal) scores with the normative 0.5 D scale. The mode was statistically significantly different between the two scales. The use of the normative scale showed slightly better agreement between the examiners and the mode, but it did not significantly decrease the variability of topographic classification among the examiners.

The significant variability in grading of corneal topography patterns by experienced topography readers reported herein highlights the challenge that faces refractive surgeons who are evaluating patients as candidates for LASIK. Although it is often easy in retrospect to find suspicious characteristics in topography maps in the eyes of patients who have developed post-LASIK ectasia, this study demonstrates why there is still some latitude in absolute determination for surgical candidacy. It is clear that not all examiners are using the same strategy for pattern evaluation, and further work on reducing the subjective variability of interpretation of corneal maps could potentially help refractive surgeons improve their ability to accurately interpret corneal topography maps. Further, the variability reported emphasizes the utility in considering a multitude of other screening parameters beyond simply the topographic pattern, whether or not it be through the use of specific screening systems such as the ERSS.

There was significant variability on the subjective topographic classifications among different examiners using the same scale. The variation in score from 0 to 4 occurred in 68% (17 of 25) of eyes with the absolute scale and in 44% (11 of 25) of the cases with the normative scale along with little agreement among the examiners, demonstrating the great variability of subjective classification of the topography.

AUTHOR CONTRIBUTIONS

Study concept and design (RA, ICR, JBR, MR, PS); data collection (RA, MRC, BMF, FPG, ICR, MR, MKS, WT); analysis and interpretation of data (RA, MWB, JOC, RC, SDK, DGD, BMF, FPG, ICR, JBR); drafting of the manuscript (MWB, ICR, JBR); critical revision of the manuscript (RA, MWB, JOC, MRC, RC, SDK, DGD, BMF, JBR, MR, MKS, PS, WT); statistical expertise (RA, SDK); administrative, technical, or material support (RA); supervision (RA)

REFERENCES

1. Rabinowitz YS, McDonnell PJ. Computer-assisted corneal topography in keratoconus. *Refract Corneal Surg.* 1989;5:400-408.
2. Wilson SE, Klyce SD. Advances in the analysis of corneal topography. *Surv Ophthalmol.* 1991;35:269-277.
3. Wilson SE, Klyce SD, Husseini ZM. Standardized color-coded maps for corneal topography. *Ophthalmology.* 1993;100:1723-1727.
4. Oshika T, Klyce SD. Corneal topography in LASIK. *Semin Ophthalmol.* 1998;13:64-70.
5. Klyce SD, Endl MJ. Corneal topography in modern refractive surgery. *Int Ophthalmol Clin.* 2002;42:19-30.
6. Smolek MK, Klyce SD, Hovis JK. The Universal Standard Scale:

proposed improvements to the American National Standards Institute (ANSI) scale for corneal topography. *Ophthalmology*. 2002;109:361-369.

7. Maeda N, Klyce SD, Smolek MK. Neural network classification of corneal topography: preliminary demonstration. *Invest Ophthalmol Vis Sci*. 1995;36:1327-1335.
8. Maeda N, Klyce SD, Smolek MK, Thompson HW. Automated keratoconus screening with corneal topography analysis. *Invest Ophthalmol Vis Sci*. 1994;35:2749-2757.
9. Rabinowitz YS. Videokeratographic indices to aid in screening for keratoconus. *J Refract Surg*. 1995;11:371-379.
10. Randleman JB, Woodward M, Lynn MJ, Stulting RD. Risk assessment for ectasia after corneal refractive surgery. *Ophthalmology*. 2008;115:37-50.
11. Randleman JB, Trattler WB, Stulting RD. Validation of the Ectasia Risk Score System for preoperative laser in situ keratomileusis screening. *Am J Ophthalmol*. 2008;145:813-818.
12. Binder PS, Trattler WB. Evaluation of a risk factor scoring system for corneal ectasia after LASIK in eyes with normal topography. *J Refract Surg*. 2010;26:241-250.
13. Spadea L, Cantera E, Cortes M, Conocchia NE, Stewart CW. Corneal ectasia after myopic laser in situ keratomileusis: a long-term study. *Clin Ophthalmol*. 2012;6:1801-1813.
14. Chan CC, Hodge C, Sutton G. External analysis of the Randleman Ectasia Risk Factor Score System: a review of 36 cases of post LASIK ectasia. *Clin Experiment Ophthalmol*. 2010;38:335-340.
15. Wilson SE, Ambrosio R. Computerized corneal topography and its importance to wavefront technology. *Cornea*. 2001;20:441-454.
16. Seiler T, Quurke AW. Iatrogenic keratectasia after LASIK in a case of forme fruste keratoconus. *J Cataract Refract Surg*. 1998;24:1007-1009.
17. Binder PS, Lindstrom RL, Stulting RD, et al. Keratoconus and corneal ectasia after LASIK. *J Refract Surg*. 2005;21:749-752.
18. Klein SR, Epstein RJ, Randleman JB, Stulting RD. Corneal ectasia after laser in situ keratomileusis in patients without apparent preoperative risk factors. *Cornea*. 2006;25:388-403.
19. Ambrósio R Jr, Dawson DG, Salomão M, Guerra FP, Caiado AL, Belin MW. Corneal ectasia after LASIK despite low preoperative risk: tomographic and biomechanical findings in the unoperated, stable, fellow eye. *J Refract Surg*. 2010;26:906-911.
20. Bühren J, Kook D, Yoon G, Kohnen T. Detection of subclinical keratoconus by using corneal anterior and posterior surface aberrations and thickness spatial profiles. *Invest Ophthalmol Vis Sci*. 2010;51:3424-3432.
21. Ambrósio R Jr, Nogueira LP, Caldas DL, et al. Evaluation of corneal shape and biomechanics before LASIK. *Int Ophthalmol Clin*. 2011;51:11-38.
22. Jafri B, Li X, Yang H, Rabinowitz YS. Higher order wavefront aberrations and topography in early and suspected keratoconus. *J Refract Surg*. 2007;23:774-781.
23. Li Y, Meisler DM, Tang M, et al. Keratoconus diagnosis with optical coherence tomography pachymetry mapping. *Ophthalmology*. 2008;115:2159-2166.
24. Ambrósio R Jr, Belin MW. Imaging of the cornea: topography vs tomography. *J Refract Surg*. 2010;26:847-849.
25. Ambrósio R Jr, Caiado AL, Guerra FP, et al. Novel pachymetric parameters based on corneal tomography for diagnosing keratoconus. *J Refract Surg*. 2011;27:753-758.
26. Ambrósio R Jr, Alonso RS, Luz A, Coca Velarde LG. Corneal-thickness spatial profile and corneal-volume distribution: tomographic indices to detect keratoconus. *J Cataract Refract Surg*. 2006;32:1851-1859.
27. Saad A, Gatinel D. Topographic and tomographic properties of forme fruste keratoconus corneas. *Invest Ophthalmol Vis Sci*. 2010;51:5546-5555.
28. Reinstein DZ, Gobbe M, Archer TJ, Silverman RH, Coleman DJ. Epithelial, stromal, and total corneal thickness in keratoconus: three-dimensional display with artemis very-high frequency digital ultrasound. *J Refract Surg*. 2010;26:259-271.
29. Reinstein DZ, Archer TJ, Gobbe M. Corneal epithelial thickness profile in the diagnosis of keratoconus. *J Refract Surg*. 2009;25:604-610.
30. Li Y, Tan O, Brass R, Weiss JL, Huang D. Corneal epithelial thickness mapping by fourier-domain optical coherence tomography in normal and keratoconic eyes. *Ophthalmology*. 2012;119:2425-2433.
31. Haque S, Simpson T, Jones L. Corneal and epithelial thickness in keratoconus: a comparison of ultrasonic pachymetry, Orbscan II, and optical coherence tomography. *J Refract Surg*. 2006;22:486-493.
32. Fontes BM, Ambrósio R Jr, Velarde GC, Nosé W. Ocular response analyzer measurements in keratoconus with normal central corneal thickness compared with matched normal control eyes. *J Refract Surg*. 2011;27:209-215.
33. Saad A, Lteif Y, Azan E, Gatinel D. Biomechanical properties of keratoconus suspect eyes. Invest *Ophthalmol Vis Sci*. 2010;51:2912-2916.
34. Ambrósio RA Jr, Randleman JB. Screening for ectasia risk: what are we screening for and how should we screen for it? *J Refract Surg*. 2013;29:230-232.
35. Amoils SP, Deist MB, Gous P, Amoils PM. Iatrogenic keratectasia after laser in situ keratomileusis for less than -4.0 to -7.0 diopters of myopia. *J Cataract Refract Surg*. 2000;26:967-977.
36. Rao SN, Epstein RJ. Early onset ectasia following laser in situ keratomileusus: case report and literature review. *J Refract Surg*. 2002;18:177-184.

APPENDIX

All Results of Topography Analysis (Classification 0, 1, 3, or 4) by the Examiners Using 1.5 and 0.5 D Scales and the Agreement With Mode in Each Case

	Examiner 1		Examiner 2		Examiner 3		Examiner 4		Examiner 5		Examiner 6		Examiner 7		Examiner 8		Examiner 9		Examiner 10		Examiner 11		Mode		Agreement With Mode (%)	
Case	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D	1.5 D	0.5 D
01	1	4	0	4	3	3	4	4	3	4	4	4	3	4	1	3	3	3	4	4	3	4	3	4	45.45	72.73
02	4	4	4	0	0	0	1	4	0	0	4	4	4	0	3	4	0	4	0	0	4	4	4	4	45.45	54.55
03	4	4	4	1	0	1	1	4	1	1	4	4	4	4	1	4	1	1	1	4	1	1	1	4	54.55	54.55
04	0	0	1	0	0	0	1	1	0	0	4	4	0	0	0	0	0	0	1	3	0	0	0	0	63.64	72.73
05	3	3	4	3	1	3	3	3	3	1	4	4	0	3	1	3	4	1	4	4	1	1	4	3	36.36	54.55
06	0	3	1	1	0	0	1	3	0	0	0	4	3	0	1	0	0	0	0	1	0	1	0	0	63.64	45.45
07	3	4	1	4	3	4	4	3	1	4	4	4	1	4	3	3	3	4	4	4	3	4	3	4	45.45	81.82
08	0	0	4	1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	90.91	72.73
09	0	1	3	3	0	1	0	1	1	1	4	4	0	4	0	1	3	0	3	3	3	3	0	1	45.45	45.45
10	0	1	1	0	0	1	0	1	0	0	4	4	0	0	0	0	3	4	0	0	0	0	0	0	72.73	54.55
11	1	1	4	3	3	3	1	3	1	3	3	3	3	3	1	3	3	3	4	3	1	3	1	3	45.45	90.91
12	0	3	1	3	3	3	0	1	0	0	4	4	1	3	0	3	3	0	3	3	0	1	0	3	45.45	54.55
13	1	4	4	4	0	4	4	4	4	4	3	4	4	4	4	4	4	3	4	4	4	4	4	4	72.73	90.91
14	0	3	4	3	0	1	3	3	3	1	3	3	3	4	3	3	3	3	4	4	0	1	3	3	54.55	54.55
15	1	3	1	4	1	1	1	3	1	3	4	3	1	4	1	3	0	3	4	4	1	1	1	3	72.73	54.55
16	4	3	1	4	0	4	3	4	3	1	4	3	1	4	4	4	3	3	4	4	3	3	4	4	36.36	54.55
17	1	3	1	4	1	1	1	3	3	1	3	3	1	3	3	3	3	1	3	4	3	1	3	3	54.55	45.45
18	0	0	0	4	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	100.00	81.82
19	1	1	1	1	0	1	1	3	1	1	3	3	1	3	3	3	0	1	1	3	1	1	1	1	63.64	54.55
20	1	3	1	3	0	1	1	3	3	1	3	3	1	3	3	3	3	0	4	4	1	3	1	3	45.45	63.64
21	0	1	1	4	0	3	1	3	1	3	0	3	0	4	0	3	0	0	1	3	0	3	0	3	63.64	63.64
22	0	1	0	3	0	0	0	3	0	0	4	4	0	0	1	1	3	0	1	0	0	1	0	0	63.64	45.45
23	0	1	0	4	0	3	0	3	0	3	0	4	0	4	0	3	0	0	1	3	0	3	0	3	90.91	54.55
24	0	0	1	1	0	1	1	3	1	0	0	1	0	0	0	1	4	0	0	0	0	0	0	0	63.64	54.55
25	0	0	0	1	0	1	1	3	1	0	0	1	0	0	0	1	4	0	0	0	1	0	0	0	63.64	54.55
Average	1	2.04	1.72	2.52	0.60	1.64	1.32	2.64	1.24	1.28	2.64	3.16	1.24	2.32	1.32	2.28	2.00	1.36	2.04	2.48	1.20	1.72	1.32	2.12	60.00	61.09
Wilcoxon signed rank (*P*)	.0007		.0502		.0003		.0002		.3899		.0332		.0166		.0010		.0701		.0191		.0207		.0033		.3544	

D = diopter